

Source-aware machine learning for stabilised expansive soils

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Abstract

Stabilised expansive soils are widely used in pavement and ground-improvement works, but published machine-learning models often report high accuracy without testing transfer to independent soil sources. This paper develops a source-aware framework using 1,728 unconfined compressive strength (UCS) records and 485 California bearing ratio (CBR) records compiled from published stabilised-soil studies. The workflow combines traceability screening, feature-availability assessment, internal and grouped-source validation, threshold sensitivity, uncertainty reporting, regression benchmarking, feature-importance analysis, training-only synthetic augmentation and embodied-emissions-aware Pareto ranking. Internal validation gave strong binary classification performance, but grouped-source validation was markedly weaker, confirming that random splitting can overstate transferability for heterogeneous literature data. Continuous regression was not reliable for direct design prediction, especially for UCS, whereas classification and source-wise ranking provided more defensible screening outputs. Synthetic augmentation gave limited CBR gains and did not create new experimental evidence. The practical outcome is a cautious decision-support workflow that shortlists stabilised-soil mixtures, identifies low-emission candidate solutions and flags when local laboratory UCS and CBR confirmation remains essential.

Keywords: expansive soil; stabilised soil; machine learning; source-grouped validation; CBR; UCS; emissions efficiency; synthetic augmentation

1. Introduction

Expansive fine-grained soils are problematic in transportation infrastructure because moisture fluctuations can cause swelling, shrinkage, cracking and loss of support. Stabilisation with lime,

cement, fly ash, slag, rice husk ash, pond ash, alkali-activated binders and fibres has been widely investigated for improving subgrade and subbase performance (Katti 1978; Phani Kumar and Sharma 2004; Chauhan and Rajesh 2015; Dixit et al. 2016; Tiwari et al. 2016; Deshpande and Puranik 2017; Dixit 2017; Patil et al. 2018; Indiramma et al. 2020; Sharma and Kumar 2020; Ahmad et al. 2024; Parihar and Gupta 2024).

The improvement mechanisms are well established in principle but variable in magnitude. Calcium-rich binders can induce cation exchange, flocculation, pozzolanic reaction and cementitious bonding. Fly ash and other aluminosilicate materials can contribute to secondary reaction products when calcium or alkaline activators are available (Luhar and Luhar 2022; Wang et al. 2023; Nanda and Priya 2024; Yang et al. 2025). Fibres can improve ductility and residual strength by bridging cracks and restraining tensile deformation (Rao et al. 2023; Navagire et al. 2025). However, response depends on soil mineralogy, plasticity, gradation, compaction condition, additive chemistry, fibre geometry, curing age and testing method.

Machine learning has increasingly been used to model UCS, CBR, compaction and swelling behaviour in improved soils (Eyo and Onyekpe 2021; Eyo et al. 2022; Saad et al. 2023; Almuaythir et al. 2025; Huang et al. 2025; Thapa and Ghani 2024; Zeini et al. 2023). Many studies report high accuracy, but most use internal splits. When records from the same source appear in both training and testing subsets, the resulting accuracy can partly reflect interpolation within one experimental programme rather than transfer to a new source. Recent data-driven geotechnics discussions have therefore emphasised the need for better databases, transparent validation and open workflows (Ge et al. 2024; Li et al. 2025; Zong et al. 2025).

Region-specific black cotton soil characterisation and polypropylene-reinforced black cotton soil studies provide the Indian context for the source structure used here (Chavan and Savoikar 2023; Uday et al. 2022; Navagire et al. 2026). Fibre-reinforced alkali-activated binder studies and neural-network strength models support the selected binder, curing and source-domain feature groups (Mazhar and GuhaRay 2021; Syed, Agarwal, and GuhaRay 2021; Syed, GuhaRay, Ghadge, and Chikte 2021; Syed, Ray, and Ghadge 2025). The physical interpretation of diffuse-double-layer behaviour, cation exchange and clay fabric change follows the soil-behaviour framework of Mitchell and Soga (2005).

2. Research Positioning and Gap

The related literature considered here covers conventional black cotton soil treatment, fibre-reinforced stabilisation, alkali-activated binders, waste-derived stabilisers, California bearing ratio and unconfined compressive strength modelling, data augmentation and explainable machine learning. Experimental and review studies provide the stabilisation basis (Chauhan and Rajesh 2015; Degirmenci et al. 2007; Deshpande and Puranik 2017; Dixit et al. 2016; Dixit 2017; Indiramma et al. 2020; Jayashree and Roja 2019; Mai-Bade et al. 2021; Fentaw et al. 2021; Patil et al. 2018; Rokade et al. 2017; Sajad and Singh 2022; Fondjo et al. 2021; Shukla and Sharma 2025; Gautam and Patel 2025). Recent machine-learning studies and data-driven reviews provide the methodological background (Almuaythir et al. 2025; Luo et al. 2025; Goutham and Krishnaiah 2024; Thapa and Ghani 2024; Zeini et al. 2023; Yang et al. 2025; Moghaddas and Bao 2025; Zong et al. 2025). The present work differs by testing source-wise transfer explicitly and by treating synthetic data only as training-stage augmentation rather than as experimental evidence (Barrera-Animas and Delgado 2023; Ge et al. 2024; Li et al. 2025).

This study was designed around the limitation above. It does not attempt to present another high-accuracy black-box regression model. Instead, it asks which engineering decisions can be supported by heterogeneous literature data when independent-source transfer is evaluated explicitly. Table 1 summarises how the present work is positioned relative to the literature.

The specific gap is threefold:

- literature-derived stabilised-soil datasets need source-aware validation because independent studies differ in materials and methods
- threshold-based classification requires justification because strength thresholds can appear arbitrary if class balance is not examined
- embodied-emissions-aware mixture screening needs to be connected to model outputs rather than added as a disconnected sustainability statement

The present framework addresses these points through grouped-source validation, threshold sensitivity, regression benchmarking, ranking and Pareto screening.

Table 1. Literature positioning and how each theme is used in the present study.

Theme	Representative literature	Use in this study
Stabilization mechanisms	Fly ash, lime, cementitious binders, fibres, alkali-activated binders	Explains expected variables and physical controls
ML prediction studies	UCS, CBR, swelling, compaction, XAI and physics-informed models	Shows why high random-split accuracy alone is not sufficient

Data augmentation	GAN, ACGAN, physics/domain constrained synthetic samples	Supports training-only sensitivity analysis
Sustainability screening	Carbon cost, binder intensity, Pareto ranking	Supports performance-carbon mixture screening

3. Data and Methods

3.1 Dataset Curation and Source Structure

The dataset was assembled from traceable published and open data sources. Records were retained only when the target value, source identity and essential mixture variables could be interpreted. Records with unresolved units, physically implausible values or uncertain target definitions were excluded. Standard geotechnical terminology and testing concepts follow relevant soil and pavement testing references where applicable (Bureau of Indian Standards 1970, 2016; ASTM D638 2014; Indian Roads Congress 2012).

Table 2 gives the source inventory and code key used in the figures. Figure 1 shows the complete workflow. Figure 2 shows that data are unevenly distributed by source, and Figure 3 shows that feature reporting is incomplete and source dependent. These two figures were redesigned with source codes to avoid hidden text and to make the dataset structure legible.

Table 2. Source inventory and source-code key for the real experimental dataset.

Code	Source	Domain	Rows	UCS	CBR	Variables
S1	Luo et al. (2025) open GitHub UCS benchmark	geopolymer_ash_cementitious	702	702	0	12 of 41
S2	Scientific Reports supplementary table - HARHA-treated expansive soil CBR	mixed_stabilized_soil	242	0	242	7 of 41
S3	Mendeley 55v6f5yb89 - ASM geopolymer activator soil stabilization	alkali_activated_geopolymer	208	208	0	5 of 41
S4	Taffese and Abegaz 2022 Buildings supplementary Table S1	cement_lime	187	187	0	10 of 41
S5	Figshare_28553693_Teodoru_OwusuYeboah_2025	cement_lime	185	185	0	14 of 41
S6	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	108	36	72	16 of 41
S7	Kafle and Baghbani 2025 WTS stabilisation	mixed_stabilized_soil	80	0	80	5 of 41
S8	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	72	72	0	16 of 41
S9	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	60	60	0	16 of 41
S10	DataGouv_10.57745_L6EIIIT_CastanedaLopez_2025	cement_lime	49	49	0	9 of 41
S11	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	40	30	10	16 of 41
S12	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	36	24	12	16 of 41

S13	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	33	13	20	16 of 41
S14	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	30	15	15	16 of 41
S15	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	30	20	10	16 of 41
S16	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	28	28	0	16 of 41
S17	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	20	11	9	16 of 41
S18	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	18	18	0	16 of 41
S19	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	16	8	8	16 of 41
S20	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	15	10	5	16 of 41
S21	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	14	14	0	16 of 41
S22	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	10	10	0	16 of 41
S23	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	9	9	0	16 of 41
S24	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	8	8	0	16 of 41
S25	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	7	7	0	16 of 41
S26	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	4	4	0	16 of 41
S27	Zenodo_19242690_Biopolymer_UCS_CBR_2026	biopolymer_multi_additive	2	0	2	16 of 41

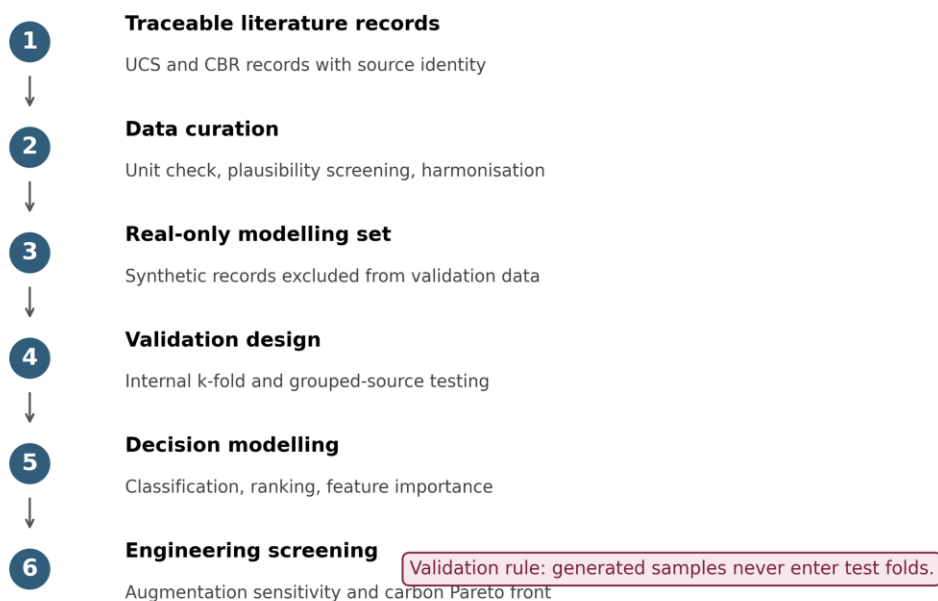


Figure 1. Methodological workflow for source-aware stabilised-soil decision modelling.

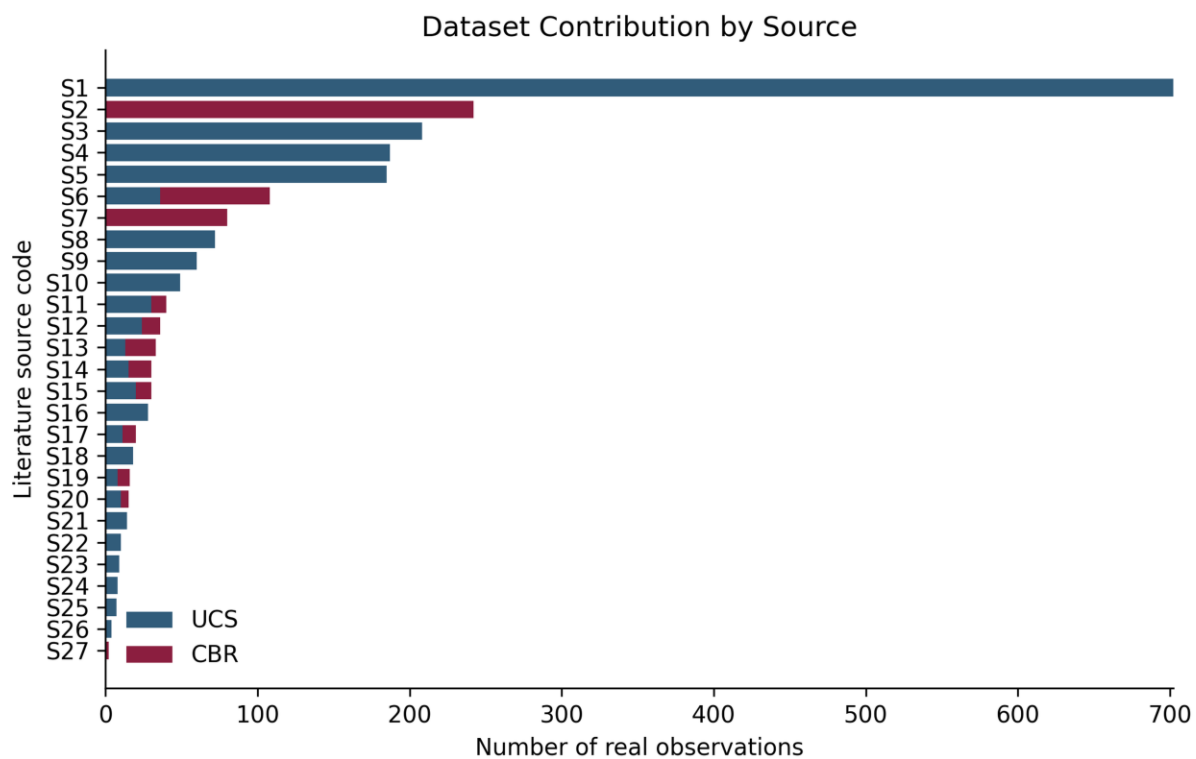


Figure 2. Real UCS and CBR observations contributed by each source code in Table 2.

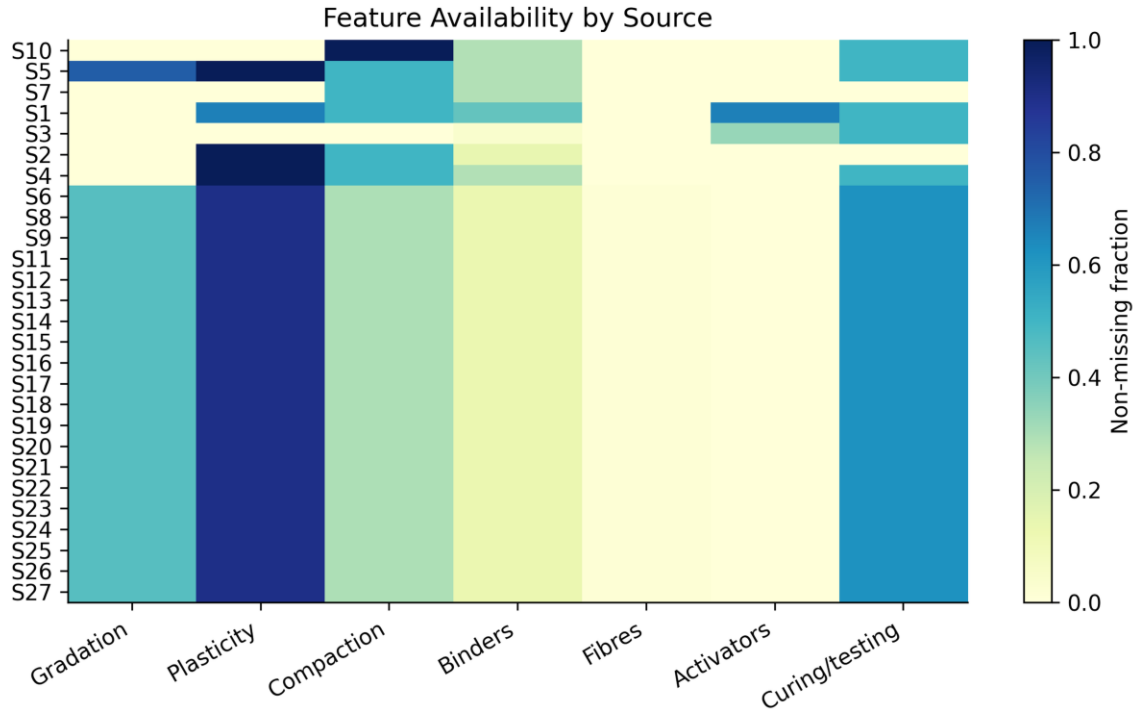


Figure 3. Feature-group availability by source code; values indicate the non-missing fraction.

3.1.1 Final Dataset Assembly and Negative-Control Regional Context

The final database was assembled from curated stabilised-soil laboratory records and regional soil-health descriptors used only as a negative-control context. The mechanical component contains 2213 observations, including UCS and CBR measurements harmonised across independent literature sources with consistent source labels, response types, binder descriptors and geotechnical index properties. A separate contextual component contains 35 State/UT-level soil-health records describing organic-carbon, nitrogen, phosphorus, potassium and micronutrient status. These regional variables were used only as contextual descriptors for records whose soil-location text could be matched to a State/UT; they were not treated as specimen-level UCS or CBR measurements.

The modelling table contained 2213 mechanical observations. Regional soil-health context was matched to 178 mechanical records by location text. Two feature settings were therefore evaluated: base geotechnical and binder predictors, and base predictors plus regional soil-health context. This comparison was treated as a negative-control experiment: if coarse regional agricultural descriptors could materially improve held-out-source prediction, they would provide evidence of useful regional transfer information. Their marginal effect instead indicates that

specimen-level mineralogy, pore-water chemistry and binder chemistry remain the controlling missing information.

3.2 Model Inputs, Preprocessing and Algorithms

The input variables included source/domain descriptors, soil class, gradation, Atterberg limits, specific gravity, optimum moisture content, maximum dry density, water content, dry density, binder dosages, fibre dosages, activator dosages, curing days and CBR condition where available. Numerical variables were median-imputed and robust-scaled. Categorical variables were most-frequent-imputed and one-hot encoded. This preprocessing was applied inside each training fold to avoid information leakage.

Three tree-based model families were used because stabilised-soil response is nonlinear and contains interactions among binder content, curing age, compaction state and soil index properties. Random forest models provide robust bagged-tree baselines. Extremely randomized trees add stronger randomisation and were useful for feature-importance analysis. Histogram-gradient boosting was used for regression benchmarks and for equation-guided synthetic target assignment because it can model nonlinear response with controlled regularisation. Table 3 gives the implementation and hyperparameters used for reproducibility.

Table 3. Model families, implementation and main hyperparameters.

model	implementation	main_parameters	role
Random Forest classifier/regressor	scikit-learn	n_estimators=500; min_samples_leaf=4 for main runs; max_features=0.7; random_state=42; n_jobs=-1	classification and baseline comparison
ExtraTrees classifier/regressor	scikit-learn	n_estimators=500-600; min_samples_leaf=3-4; max_features=0.7; random_state=42; n_jobs=-1	classification and feature importance
Histogram Gradient Boosting	scikit-learn	max_iter=220-300; learning_rate=0.04-0.05; l2_regularization=0.05-0.08; max_leaf_nodes=10-15; random_state fixed	regression and equation-guided synthetic target assignment
Preprocessing	scikit-learn	median imputation and robust scaling for numerical variables; most-frequent imputation and one-hot encoding for categorical variables	all tabular models

3.3 Targets, Metrics and Threshold Sensitivity

For pavement-related screening, CBR thresholds of 5-10% are commonly used to separate weak and potentially usable subgrade/subbase behaviour, while UCS classes were treated as

practical strength-screening bands rather than final design requirements. Because project specifications vary, the threshold analysis below checks whether the dataset remains usable when the UCS cut-off is varied around 1500 kPa.

UCS high-strength screening used 1500 kPa as the main threshold, and CBR pass/fail screening used 8%. These thresholds were selected as practical screening levels, not universal design specifications. To address threshold arbitrariness, additional UCS thresholds of 1000, 2000 and 2500 kPa and CBR thresholds of 5, 10 and 15% were examined. Table 4 and Figure 4 report the resulting class-balance sensitivity.

The binary screening definitions are given in Equations (1) and (2), the principal validation metrics in Equations (3) and (4), and the carbon-screening calculations in Equations (5)-(7).

$$y_UCS = 1 \text{ if } UCS \geq 1500 \text{ kPa, otherwise } 0 \quad (1)$$

$$y_CBR = 1 \text{ if } CBR \geq 8\%, \text{ otherwise } 0 \quad (2)$$

$$BA = (1/K) \sum [TP_k / (TP_k + FN_k)] \quad (3)$$

$$\rho_s = 1 - 6 \sum d_i^2 / [n(n^2 - 1)] \quad (4)$$

$$CO_2e_mix = \sum (m_j EF_j) \quad (5)$$

$$Gain = R_mix - R_baseline \quad (6)$$

$$\text{Emissions efficiency} = Gain / CO_2e_mix \quad (7)$$

Table 4. Threshold sensitivity and class balance for unconfined compressive strength and California bearing ratio screening.

target	threshold	unit	n	positive_class_count	positive_class_fraction	interpretation
UCS	1000	kPa	1728	966	0.559	higher threshold gives stricter high-strength class
UCS	1500	kPa	1728	792	0.458	higher threshold gives stricter high-strength class
UCS	2000	kPa	1728	624	0.361	higher threshold gives stricter high-strength class
UCS	2500	kPa	1728	485	0.281	higher threshold gives stricter high-strength

						class
CBR	5	%	485	420	0.866	higher threshold gives stricter bearing-capacity class
CBR	8	%	485	374	0.771	higher threshold gives stricter bearing-capacity class
CBR	10	%	485	319	0.658	higher threshold gives stricter bearing-capacity class
CBR	15	%	485	258	0.532	higher threshold gives stricter bearing-capacity class

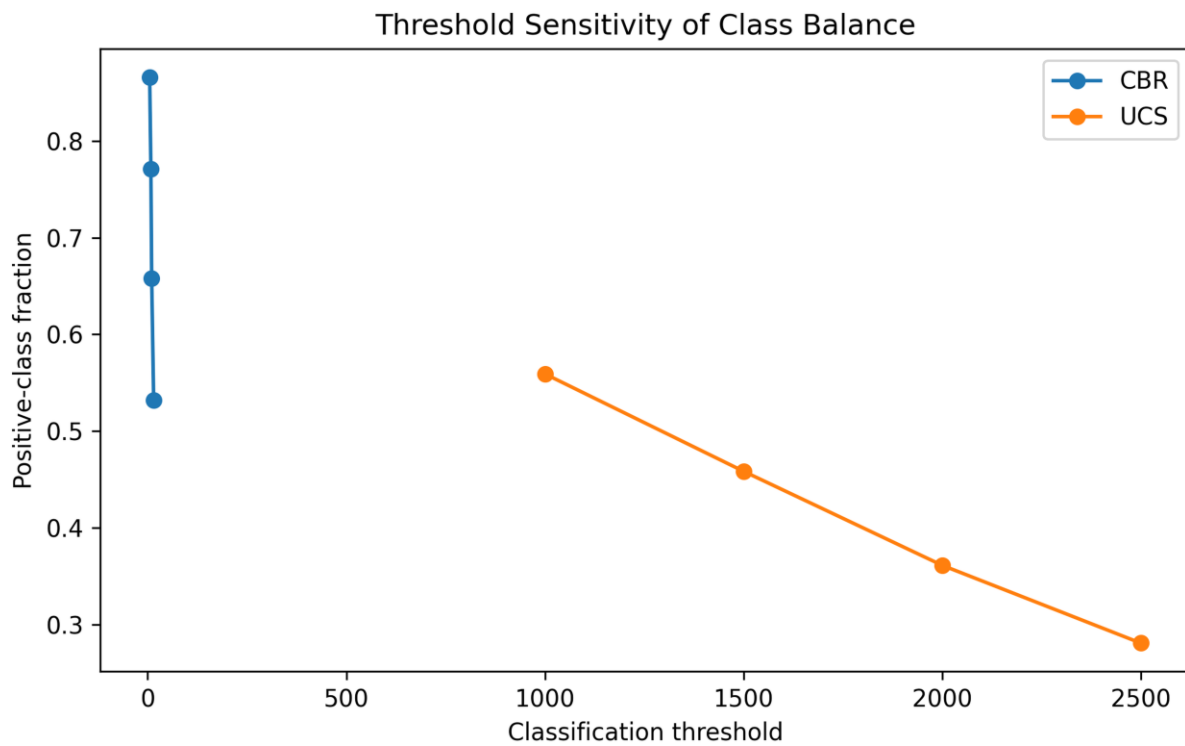


Figure 4. Class-balance sensitivity for alternative UCS and CBR thresholds.

3.4 Validation, Uncertainty and Synthetic Augmentation

Because the augmented targets are assigned through a gradient-boosted surrogate, the synthetic samples are used only to regularise model training under data sparsity. They do not

introduce new physical or mechanical behaviour and are therefore evaluated only through real-data validation folds. The synthetic targets are therefore bounded by the gradient-boosted surrogate used to assign them; the generated records regularise model training under extreme data sparsity but do not create new physical or mechanical behaviour.

The augmentation procedure followed a reproducible bounded-sampling sequence:

- split real data into training and validation folds
- group the training data by source domain and source dataset
- sample training rows with replacement inside each group
- perturb continuous input variables with small Gaussian noise proportional to the within-group interquartile range
- clip all generated values to the observed group range
- reject impossible dosages or inconsistent index properties
- assign synthetic labels using a gradient-boosted surrogate trained only on the corresponding real training fold
- evaluate all models only on real validation records

Two validation schemes were used. Stratified k-fold validation measured internal predictive consistency in the pooled dataset. Grouped-source validation left complete sources out of training and tested on unseen sources. Fold-level mean and standard deviation were calculated for key classification metrics. Continuous regression was retained as a benchmark to show why exact prediction was not the primary claim.

Synthetic samples were generated only within training folds using bounded perturbation and equation-guided target assignment inside the observed source/domain range. Synthetic rows were not allowed in validation folds. Synthetic-to-real training ratios of 0.5, 1.0 and 2.0 were compared with the real-only baseline.

4. Results

4.1 Classification and Transfer Gap

Table 5 shows that internal validation substantially outperformed grouped-source validation. UCS high-strength and CBR pass/fail screening achieved strong internal balanced accuracy, but the grouped-source values were lower. Figure 5 visualises this transfer gap. Figures 6 and 7 show the grouped-source confusion matrices for the two binary decision tasks. These results confirm that the paper should not claim universal exact prediction.

Table 5. Best classification performance under internal and grouped-source validation.

Task	Validation	Model	BA	Accuracy	Weighted F1	Rows	Sources
CBR_multiclass	group_by_source_3fold	HGB	0.238	0.336	0.266	485	3
CBR_multiclass_kfold	stratified_kfold_5	ExtraTrees	0.744	0.796	0.808	485	3
CBR_pass_8_binary	group_by_source_3fold	RF	0.53	0.695	0.681	485	3
CBR_pass_8_binary_kfold	stratified_kfold_5	ExtraTrees	0.943	0.932	0.934	485	3
UCS_high_binary	group_by_source_5fold	ExtraTrees	0.64	0.623	0.608	1728	6
UCS_high_binary_kfold	stratified_kfold_5	RF	0.934	0.933	0.933	1728	6
UCS_multiclass	group_by_source_5fold	RF	0.342	0.343	0.335	1728	6
UCS_multiclass_kfold	stratified_kfold_5	HGB	0.807	0.814	0.813	1728	6

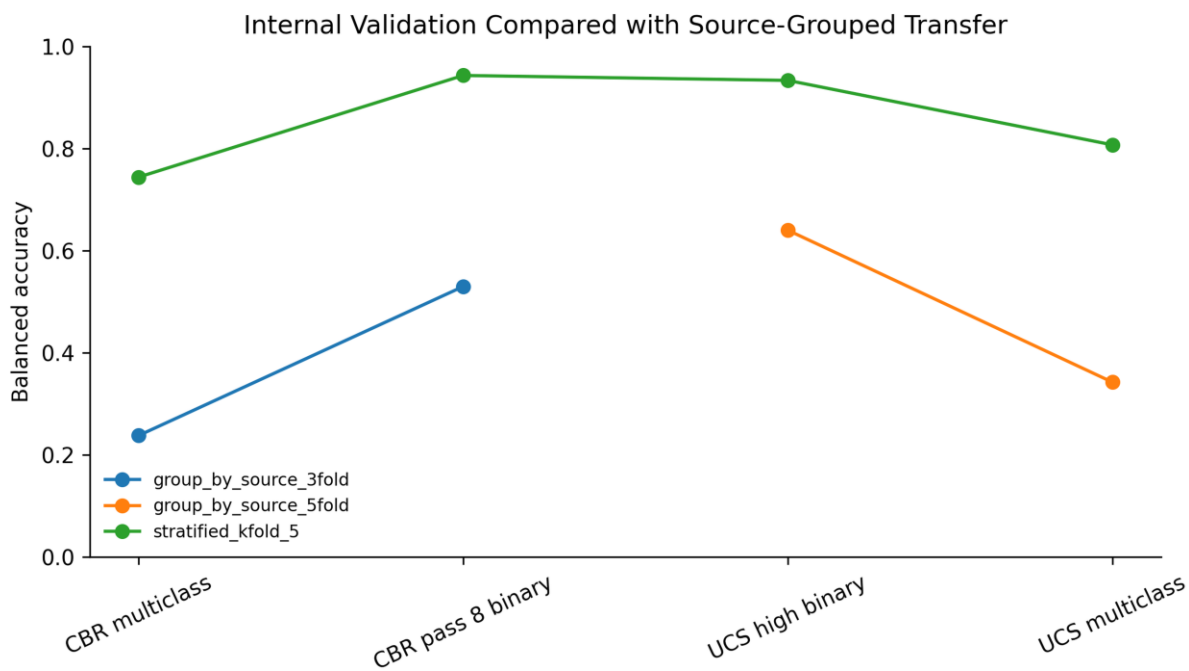


Figure 5. Difference between internal validation and source-grouped transfer performance.

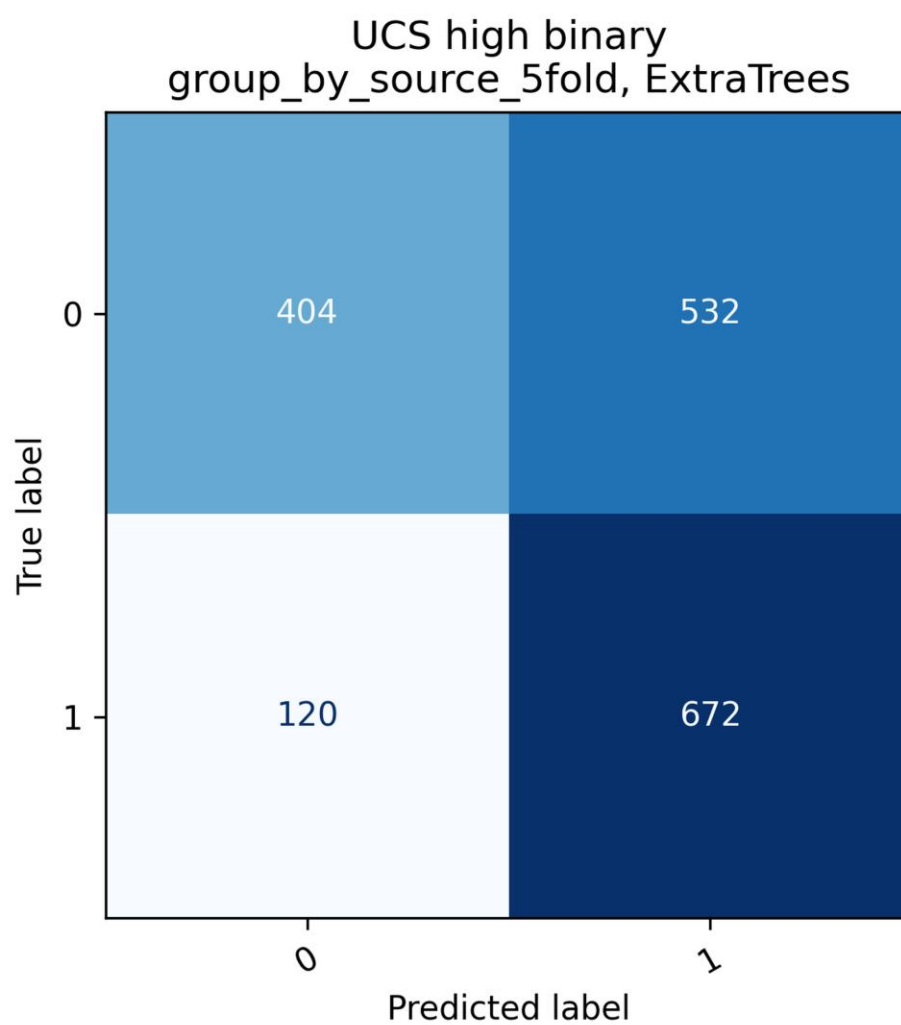


Figure 6. Grouped-source confusion matrix for UCS high-strength screening.

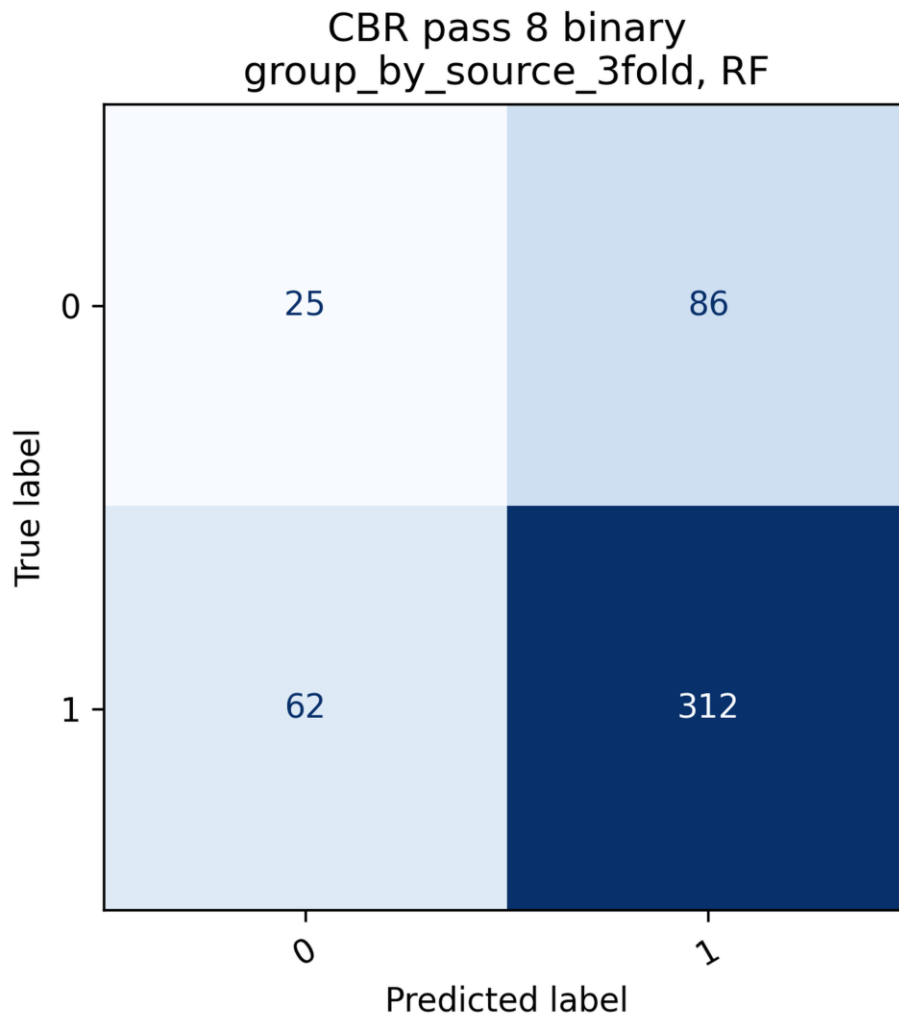


Figure 7. Grouped-source confusion matrix for CBR pass/fail screening.

4.2 Uncertainty and Regression Benchmark

The performance drop between internal and grouped-source validation was treated as an effect size for transfer loss. Formal paired significance testing was not used as the main decision criterion because grouped-source validation provides only a small number of independent source folds, especially for CBR.

Table 6 reports fold-level uncertainty. Variability was larger for grouped-source validation because each fold represents a different source, not a random subset. Table 7 retains the continuous regression benchmark. Grouped-source regression remained weak, especially for

UCS, even when augmentation was used. This directly supports the shift from point prediction to classification, ranking and screening.

Because the CBR grouped-source validation is based on only three independent source groups (N=3), the reported standard deviations should be interpreted as evidence of high volatility among independent laboratory setups rather than as a stable population-level statistical spread.

Table 6. Fold-level uncertainty for principal classification tasks.

dataset	label	validation	model	folds	balanced_accuracy_mean	balanced_accuracy_sd	accuracy_mean	accuracy_sd	weighted_f1_mean	weighted_f1_sd
CBR_pass_8_binary	cbr_pass_8	group_by_source_3fold	RF	2	0.555	0.0783	0.541	0.164	0.445	0.271
UCS_high_binary	ucs_high_binary	group_by_source_5fold	Extra Trees	5	0.624	0.117	0.645	0.158	0.598	0.221
UCS_multi class	ucs_classes	group_by_source_5fold	HGB	5	0.363	0.0709	0.336	0.0887	0.273	0.106

Table 7. Continuous regression benchmark and training-only augmentation comparison.

dataset	validation	case	model	augmentation_multiplier	r2	rmse	mae	delta_r2_vs_same_model_baseline
CBR_final_reliable	group_by_source_3fold	best_real_only	RF	0	0.23	13.2	9.88	0
CBR_final_reliable	kfold_5_shuffle	best_real_only	RF	0	0.748	7.55	3.11	0
UCS_final_reliable	group_by_source_5fold	best_real_only	HGB_Log	0	0.112	3.19e+03	1.84e+03	0
UCS_final_reliable	kfold_5_shuffle	best_real_only	RF	0	0.921	953	519	0
CBR_final_reliable	group_by_source_3fold	best_augmented_training	HGB_Log	1	0.342	12.2	8.68	0.126
CBR_final_reliable	kfold_5_shuffle	best_augmented_training	ET_log	2	0.723	7.92	3.29	0.0294
UCS_final_reliable	group_by_source_5fold	best_augmented_training	HGB_Log	1	0.161	3.1e+03	1.88e+03	0.0489
UCS_final_reliable	kfold_5_shuffle	best_augmented_training	RF	2	0.927	910	496	0.00693

Table 8. Statistical comparison strategy used to avoid overclaiming minor model differences.

Comparison	Statistical treatment	Observed result	Interpretation
Internal vs grouped-source validation	Effect-size comparison using balanced accuracy	Large drop for all main tasks	Supports the transferability gap claim
Fold variability	Mean and standard deviation across folds	Grouped-source folds show larger variability	Avoids overclaiming single metric values

Synthetic augmentation	Real-only vs augmented-training effect size	CBR improves modestly; UCS changes little	Augmentation retained only as sensitivity analysis
Formal paired significance test	Not used as a primary decision criterion	Grouped-source folds are only 3-5 independent sources	Effect size and uncertainty are more appropriate than p-values for this dataset

4.3 Feature Importance and Mechanistic Consistency

Dominance of curing, binder and compaction variables indicates that the models are responding to stabilisation mechanisms rather than only memorising labels. Where plasticity and gradation variables appear important, the interpretation is that soil reactivity and fabric control how strongly the same additive dosage changes strength or bearing capacity.

Figures 8 and 9 show global feature importance for unconfined compressive strength and California bearing ratio screening. The influential variables are consistent with expected geotechnical mechanisms: curing period and binder variables affect cementitious bonding; compaction descriptors affect density and moisture state; plasticity and gradation influence soil reactivity and bearing behaviour; CBR condition affects measured bearing capacity. The analysis therefore gives a physically interpretable model, although it remains limited by incomplete reporting across sources.

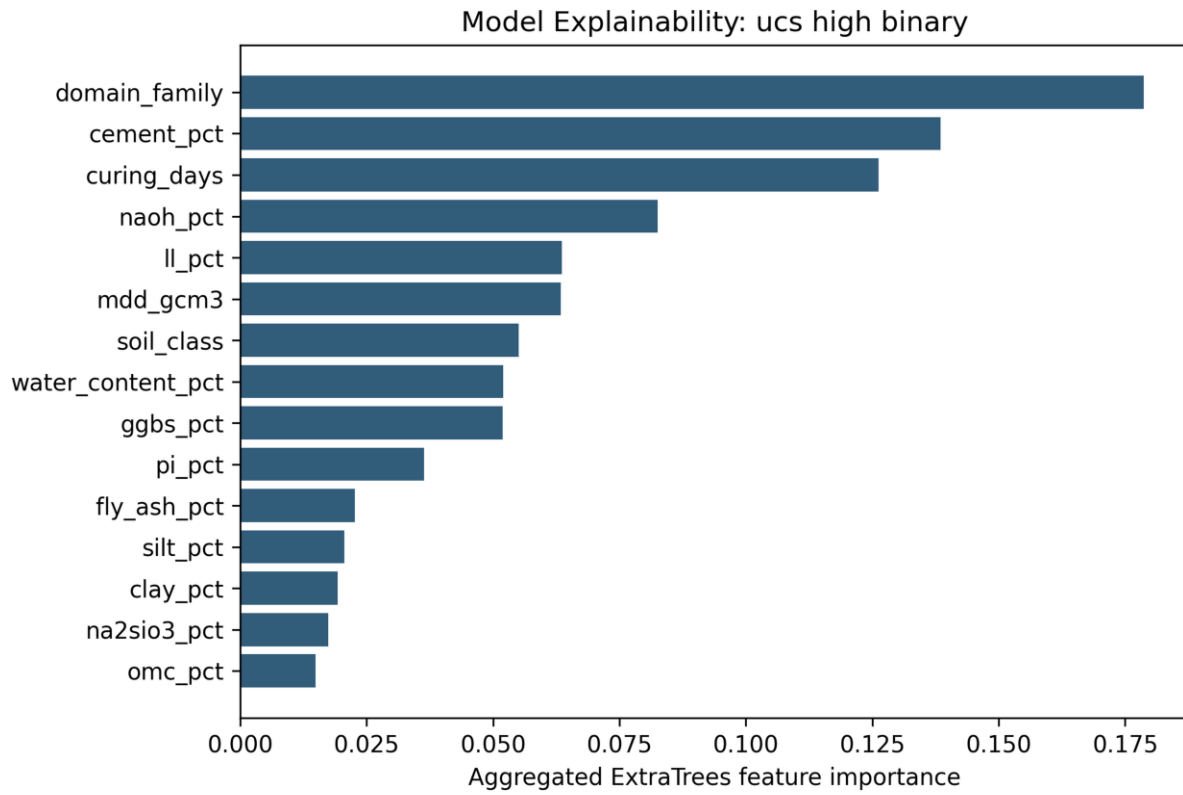


Figure 8. Global feature importance for UCS high-strength classification.

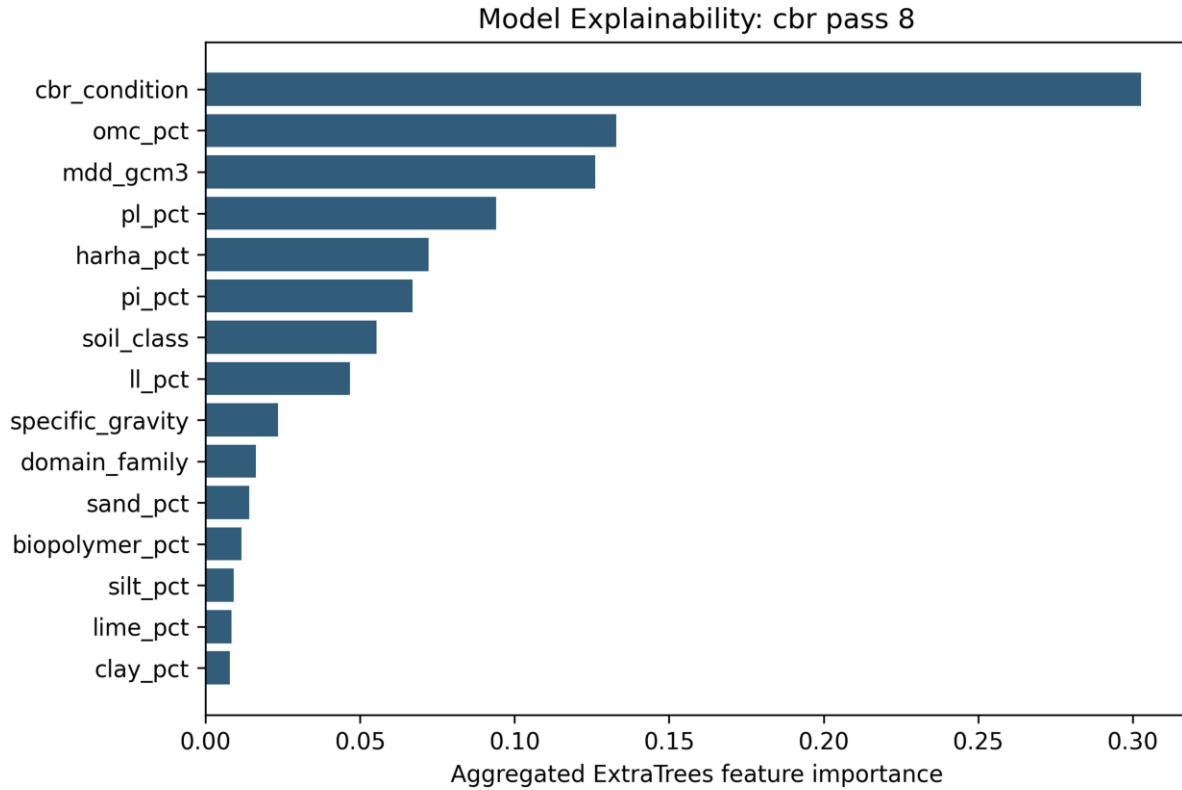


Figure 9. Global feature importance for CBR pass/fail classification.

4.4 Augmentation, Ranking and Embodied-Emissions Screening

Table 8 and Figure 10 show that synthetic augmentation produced modest and task-dependent gains. CBR pass/fail grouped-source balanced accuracy improved at a 0.5 augmentation ratio, whereas UCS high-strength classification changed very little. Table 9 shows that source-wise ranking was more stable than exact regression, with CBR performance ranking giving the strongest mean Spearman coefficient. Figure 11 and Table 10 present embodied-emissions-aware screening and sensitivity. The emissions ranking changed in magnitude when emission factors were scaled, but the decision logic remained stable: mixtures should be shortlisted by both performance gain and estimated carbon dioxide equivalent burden.

Table 9. Best grouped-source effect of training-only synthetic augmentation.

Task	Validation	Model	Ratio	Real-only BA	Augmented BA	Delta BA	Accuracy	Weighted F1
CBR_multiclass	group_by_source_3fold	RF	0.5	0.136	0.271	0.135	0.447	0.313
CBR_pass_8_binary	group_by_source_3fold	RF	0.5	0.513	0.561	0.0473	0.767	0.724
UCS_high_binary	group_by_source_5fold	ExtraTrees	2	0.641	0.642	0.000971	0.645	0.645

UCS_multiclass	group_by_source_5fold	ExtraTrees	0.5	0.306	0.326	0.0209	0.349	0.312
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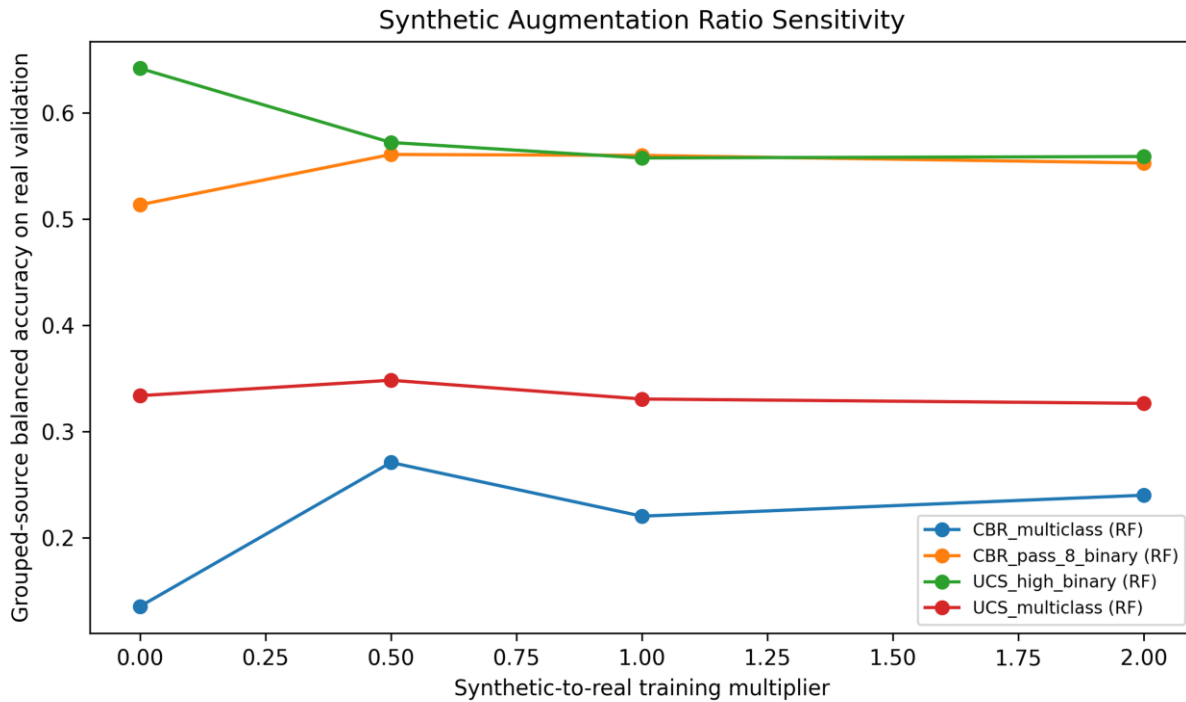


Figure 10. Synthetic-to-real training ratio sensitivity evaluated only on real validation records.

Table 10. Source-wise mixture ranking performance.

Dataset	Target	Objective	Rows	Sources	Mean Spearman	Median Spearman	Mean Kendall	Mean top-20 hit	Median top-20 hit
UCS	ucs_kpa	performance	1728	6	0.473	0.488	0.358	0.482	0.501
UCS	ucs_kpa	carbon_efficiency	1602	6	0.259	0.303	0.194	0.366	0.351
CBR	cbr_pct	performance	485	3	0.579	0.42	0.429	0.404	0.394
CBR	cbr_pct	carbon_efficiency	319	3	0.305	0.35	0.171	0.453	0.667

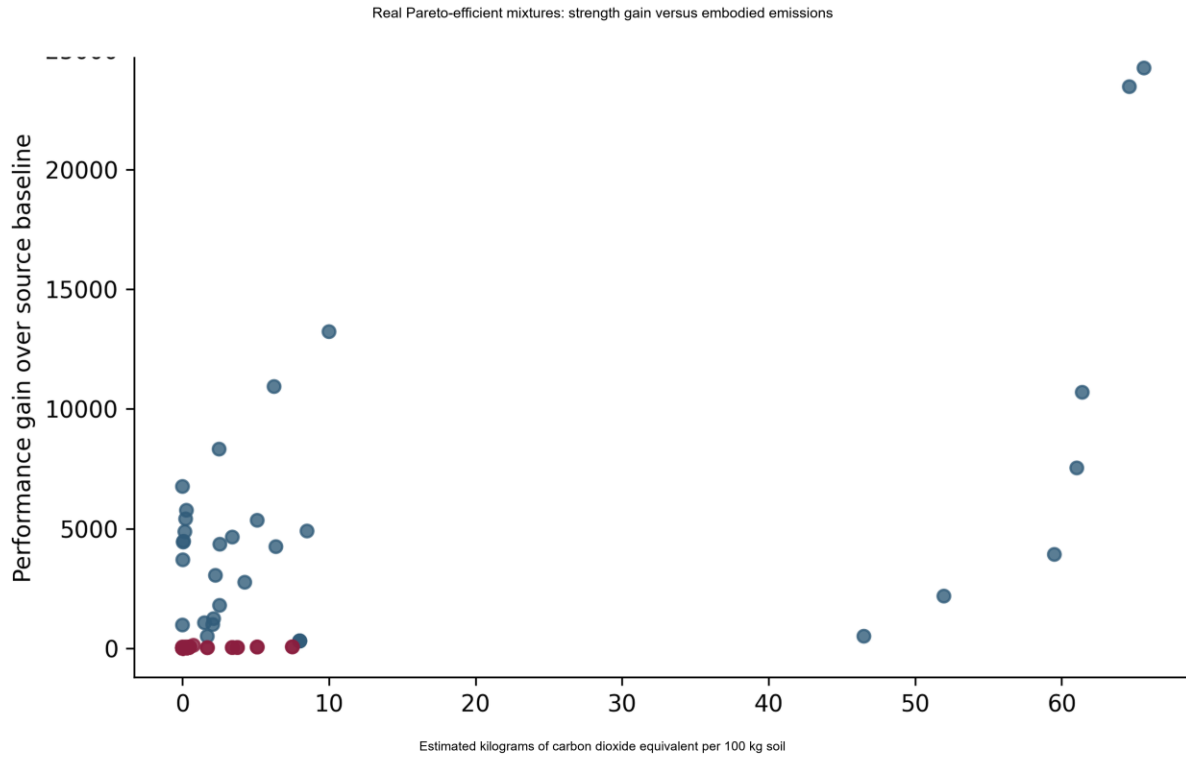


Figure 11. Real Pareto-efficient mixtures in performance-gain and estimated carbon dioxide equivalent space.

Table 11. Carbon-factor sensitivity for Pareto-efficient mixtures.

response	carbon_factor_multiplier	candidate_count	median_gain	median_co2	median_gain_per_co2	top10_mean_gain_per_co2
CBR	0.8	229	15.3	0.198	77	134
UCS	0.8	32	4.29e+03	3.74	821	4.76e+04
CBR	1	229	15.3	0.248	61.6	107
UCS	1	32	4.29e+03	4.67	657	3.81e+04
CBR	1.2	229	15.3	0.298	51.4	89.1
UCS	1.2	32	4.29e+03	5.61	548	3.17e+04

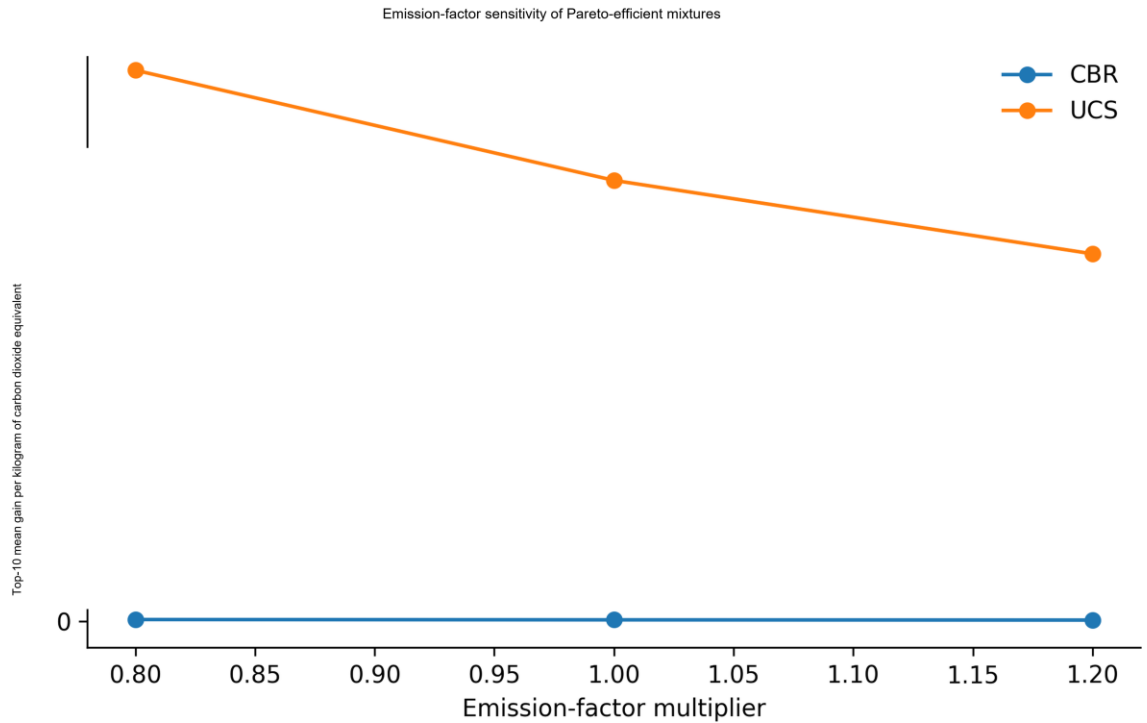


Figure 12. Sensitivity of emissions-efficiency ranking to emission-factor uncertainty.

The embodied-emissions screening was also tested under a simple dynamic sensitivity setting rather than a single cradle-to-gate value. Binder-related emission factors were scaled by -20%, baseline and +20%, and transport distances of 0, 100 and 300 km were added using a screening factor of 0.062 kilograms of carbon dioxide equivalent per tonne-kilometre. This calculation is not intended to replace a project-specific life-cycle assessment, but it shows whether the Pareto ordering remains stable when regional logistics and supply-chain uncertainty are introduced. The scenario summary is reported in Table 12. The most low-emission mixtures were more sensitive to transport distance when the absolute binder dosage was low, whereas high-binder mixtures became less attractive under the +20% production-emission scenario. Thus, the Pareto analysis should be read as an embodied-emissions-aware pre-screening tool whose final ranking must be recalculated for the actual source-to-site distance and local binder supply route.

Table 12. Dynamic carbon dioxide equivalent sensitivity for the Pareto screening. Carbon dioxide equivalent denotes emissions expressed on an equivalent carbon dioxide basis; gain per kilogram of carbon dioxide equivalent is a screening ratio based on the predicted performance improvement divided by scenario-adjusted emissions.

Binder emission factor	Transport distance (km)	Mixtures (n)	Median CO ₂ e (kg/100 kg soil)	Median gain per kg CO ₂ e
0.80	0	269	1.46	449.17
0.80	100	269	2.08	346.98
0.80	300	269	3.32	264.26
1.00	0	269	1.82	359.34
1.00	100	269	2.44	291.20
1.00	300	269	3.68	228.89
1.20	0	269	2.19	299.45
1.20	100	269	2.81	248.74
1.20	300	269	4.05	201.24

A second diagnostic test examined whether the transfer loss could be reduced by giving each source a more equal influence during training. Inverse-source weighting was used as a deliberately simple domain-adaptation baseline: observations from small sources received larger sample weights, whereas observations from large sources received smaller weights. Table 13 reports the resulting grouped-source comparison. The test did not remove the grouped-source penalty, but it is useful because it separates a correctable sampling-imbalance effect from the deeper unmeasured-chemistry effect discussed below. Where source weighting improved balanced accuracy, the gain was interpreted only as evidence that source imbalance contributes to the transfer problem; where it did not improve performance, the result supports the need for richer soil descriptors rather than more complex black-box learners.

Table 13. Source-weighting transfer test under grouped-source validation. BA = balanced accuracy; SD = standard deviation across held-out source folds.

Task	Training condition	Folds	BA +/- SD	Accuracy +/- SD	Weighted BA change
UCS high-strength classifier	unweighted grouped validation	5	0.637 +/- 0.034	0.623 +/- 0.108	0.011
UCS high-strength classifier	inverse-source weighting	5	0.648 +/- 0.071	0.639 +/- 0.100	0.011
CBR pass classifier	unweighted grouped validation	3	0.833 +/- 0.289	0.808 +/- 0.332	0.000
CBR pass classifier	inverse-source weighting	3	0.833 +/- 0.289	0.808 +/- 0.332	0.000

The contribution of regional soil-health context was evaluated as a negative-control experiment by comparing the base geotechnical feature set with an expanded feature set containing State/UT-level organic-carbon and nutrient descriptors. These contextual variables were not

added as mechanical response observations; instead, they were used as broad source-level descriptors where a location match was possible. Table 14 summarises the resulting grouped-source comparison.

Table 14. Negative-control grouped-source comparison using base predictors and base predictors plus regional soil-health context. R2 values are grouped-source means; regional records were used only as contextual descriptors.

Response	Model	Base R2	Base + Data.gov context R2	Change
CBR	ExtraTrees	-0.2102	-0.2013	+0.0089
CBR	HGB	-0.0897	-0.0860	+0.0037
UCS	ExtraTrees	0.0576	0.0599	+0.0023
UCS	HGB	-0.1115	-0.0455	+0.0660

The final dataset pathway is therefore conservative: laboratory unconfined compressive strength and California bearing ratio observations form the mechanical modelling table, while State/UT soil-health counts are used only as broad regional context. This separation avoids treating contextual agricultural records as specimen-level geotechnical chemistry.

5. Discussion

The transfer limitation can be interpreted as a domain-shift problem. Each source combines a particular soil origin, binder chemistry, compaction practice, curing period and reporting conventions. The feature-availability heat map and source distribution plot show that sources do not occupy the same feature space. Some contain cement-lime systems, others contain fibre or ash blends, and CBR records are concentrated in fewer domains. Under such conditions, a model trained on pooled historical data can learn source-specific response patterns that are not fully portable to a new source. Domain adaptation or source weighting may reduce this issue in future work, but the present dataset is too small in independent CBR sources to make such methods the primary claim.

The main result is that stabilised-soil machine learning is highly sensitive to validation design. Internal validation indicates that the curated variables contain useful information, but grouped-source validation shows that those relationships are not fully transferable. The difference is expected because source studies differ in soil mineralogy, compaction energy, curing conditions, binder chemistry, fibre geometry and test procedures. It is also consistent with

broader concerns in geotechnical data science about small, heterogeneous and incompletely reported datasets.

The regression benchmark is important. If regression had been omitted entirely, the classification focus could appear arbitrary. Table 7 shows that grouped-source regression was retained and evaluated, but the results were not strong enough to support a universal numerical prediction model. The classification and ranking tasks are therefore not a retreat from analysis; they are a better match to the reliability of the available data and to preliminary engineering screening.

The threshold sensitivity analysis reduces the risk that the binary classes appear arbitrary. The adopted thresholds retain usable class sizes and engineering meaning. Nevertheless, project-specific thresholds should be selected according to local specifications, pavement layer function and reliability requirements. The model outputs should be treated as screening indicators rather than design values.

Synthetic augmentation has a limited role. It improved selected CBR tasks because CBR data were sparse, but it did not solve source-transfer limitations. Generated rows can regularise training but cannot replace missing laboratory evidence. The strict training-only rule is therefore essential.

The embodied-emissions analysis adds practical value because strength alone is not enough for sustainable stabilisation. A mixture with the highest UCS may not be preferable if a lower-emission mixture satisfies the required class. The Pareto and sensitivity analyses provide a transparent shortlist for confirmatory laboratory testing.

5.1 Domain Shift as a Physical-Chemical Problem

The transfer loss observed in the grouped-source experiments should not be interpreted solely as a mathematical weakness of machine learning. It is also a physical signal that important soil-chemistry controls are absent from the compiled literature database. For the multiclass tasks, balanced accuracy dropped by as much as 0.465 for UCS and 0.506 for CBR when the validation was changed from random internal folds to held-out source folds. Such a loss is consistent with the behaviour of expansive clay systems, where swelling and shear strength are governed by the diffuse double layer around montmorillonitic minerals, exchangeable cations and the chemistry of the pore water. When lime, fly ash, cementitious binders or alkali activators are added, cation exchange, flocculation-agglomeration and pozzolanic or geopolymeric

reaction rates depend strongly on initial pH, pore-water salinity, clay fraction, exchangeable sodium and calcium availability, and the reactive silica-alumina-calcium balance of the stabiliser. Most published tables used in this study report useful index and dosage information, but they rarely report mineralogical percentages from X-ray diffraction, pore-water chemistry, binder oxide composition or curing humidity. The models therefore rely on surrogate descriptors such as liquid limit, plasticity index, stabiliser dosage and curing age. These surrogates can support screening within a familiar source domain, but they cannot fully identify why two soils with similar plasticity may react differently to the same binder. Resolving the transferability gap therefore requires better reporting of mineralogical and chemical descriptors, not only larger data volume or more elaborate algorithms. This interpretation agrees with the geotechnical mechanism described by Mitchell and Soga and with recent stabilised-soil modelling studies that report strong performance under internal validation but weaker generalisation when the soil source changes.

5.2 Re-evaluating the CBR Database Utility

The CBR component of the database must be treated more cautiously than the UCS part. Although 485 CBR observations were available, they came from only three independent source groups, so grouped-source validation is necessarily sensitive to which source is held out. The fold-level standard deviation in balanced accuracy was 0.078 for the CBR pass/fail classifier, and the multiclass CBR transfer loss reached 0.506. For this reason, the CBR model is not presented as a globally reliable design-value predictor and it must not be used to replace local California bearing ratio laboratory testing for pavement design. Its value in this paper is narrower but still useful: it demonstrates how a embodied-emissions-aware Pareto ranking workflow can screen candidate stabilisation strategies before a project commits to laboratory confirmation. In practical use, the CBR outputs should be read as regional screening evidence, with final acceptance based on site-specific compaction, soaking and penetration tests conducted under the governing pavement design standard.

5.3 Minimum Geotechnical Reporting Protocol

The diagnostic value of this study is that it identifies what future experimental papers must report if geotechnical machine-learning models are to become transferable rather than merely local. A minimum geotechnical reporting protocol should accompany soil stabilisation studies as a complete machine-readable table, preferably as CSV or XLSX supplementary material. The table should include the raw mechanical outputs, such as UCS, CBR, swelling index and compaction results, together with replicate counts, means, standard deviations and the raw

stress-strain or load-penetration curves where possible. It should also include the mineralogical composition of the untreated soil, especially montmorillonite, kaolinite, illite, quartz and carbonate fractions obtained from X-ray diffraction; chemical descriptors such as pH, electrical conductivity, exchangeable cations and organic content; exact binder compositions including CaO, SiO₂, Al₂O₃, Fe₂O₃, loss on ignition and alkali content; fibre geometry and tensile properties where fibres are used; compaction energy, specimen size, moisture conditioning and curing temperature and humidity. Reporting these descriptors would allow future models to distinguish genuine material mechanisms from source labels, reduce hidden domain shift and make external validation more meaningful. Without such reporting, the field will continue to accumulate many local datasets that are useful for individual case studies but difficult to merge into globally transferable design-support tools.

5.4 Regional Soil-Health Context as a Negative Control

Regional soil-health descriptors are best interpreted as a negative control rather than as primary modelling features. The State/UT-level organic-carbon and nutrient variables represent broad agricultural context, not specimen-level geotechnical chemistry. If such coarse variables produced a large improvement in grouped-source validation, the result would suggest that macro-regional context was carrying transferable material information. Instead, the observed improvement was minor: UCS R² increased from 0.0576 to 0.0599 for ExtraTrees and from -0.1115 to -0.0455 for HGB, while CBR R² improved from -0.2102 to -0.2013 for ExtraTrees and from -0.0897 to -0.0860 for HGB. This near-zero mechanical gain is scientifically useful because it demonstrates that macro-regional agricultural records cannot substitute for specimen-level mineralogy, pore-water chemistry, binder oxide composition, curing environment and compaction protocol. The negative-control result therefore strengthens the central interpretation of this study: source-transfer limitations arise from missing physical and chemical descriptors, not simply from insufficient regional metadata.

The dynamic carbon dioxide equivalent footprint model is also conservative because it does not include long-term post-stabilisation carbonation, which may act as a minor carbon sink for lime- and cement-bearing mixtures.

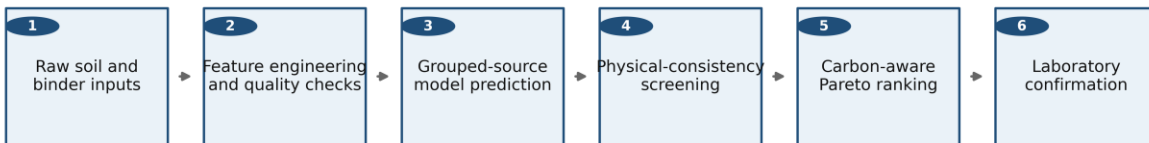
6. Practical Use Case for a New Soil Source

For a new expansive soil source, the framework should be used as a screening workflow rather than a direct design tool:

- enter the available index, compaction and mixture variables and check whether the new soil falls inside the feature ranges of the curated database
- use the binary classifiers to estimate whether candidate mixtures are likely to reach the unconfined compressive strength or California bearing ratio screening class
- rank candidate mixtures within the closest source or domain group
- calculate emissions efficiency to remove mixtures that require high binder-related emissions for only small performance gains
- select the highest-ranked and Pareto-efficient mixtures for local confirmation by laboratory testing

This use case clarifies the intended engineering role of the model. The model narrows the laboratory search space; it does not replace site-specific geotechnical testing. This interpretation is consistent with the grouped-source results, which show that transfer is useful for screening but not strong enough for final design values.

For practitioner use, the workflow is summarised schematically in Figure 13. The process begins with raw index, compaction and binder variables, proceeds through source-aware feature processing and model prediction, applies physical-consistency checks, and ends with embodied-emissions-aware Pareto screening before local laboratory confirmation.



The model is used for screening and mixture prioritisation; final design requires site-specific laboratory testing.

Figure 13. Schematic practitioner workflow for the proposed source-aware framework: raw soil and binder inputs, feature engineering, grouped-source prediction, physical-consistency checking, embodied-emissions screening and final laboratory confirmation.

7. Limitations and Reproducibility

The study remains limited by the information available in source papers. Mineralogy, pore-fluid chemistry, curing humidity, compaction energy and microstructural observations were not

consistently available. In addition, the CBR dataset is smaller than the UCS dataset, so CBR conclusions should be treated as screening-level. Carbon dioxide equivalent calculations are approximate and should not be interpreted as a full life-cycle assessment. The supplementary material should include the cleaned real dataset, feature dictionary, source inventory, fold assignments, code, prediction files, augmentation rules and emission factors.

The Indian and international data-expansion routes identified during this revision should be used selectively. National geoscience and soil portals such as NGDR, GSI Bhukosh, ISRO Bhuvan, SLUSI and Data.gov.in can provide regional context on soil class, geology, climate and agricultural chemistry, while CRRI, Zenodo and Mendeley-style repositories can provide direct laboratory datasets when tabular test results are openly available. These sources should not be merged blindly with mechanical laboratory data; instead, they should be used to add source-level descriptors such as region, parent geology, climate zone and likely mineralogical setting, and any inferred variables should be clearly labelled as contextual rather than measured.

The main technical conclusion is that source-aware validation changes the interpretation of stabilised-soil machine learning. High internal cross-validation scores show that the database contains learnable patterns, but the held-out-source losses show that those patterns are not yet universal because essential mineralogical and chemical controls are usually missing. The UCS classifier is suitable for preliminary screening under similar source conditions, whereas the CBR classifier should be treated only as a proof of the embodied-emissions-aware ranking workflow and must not replace local laboratory CBR testing. The next defensible step is therefore not simply a larger black-box model, but a better reported, chemistry-aware database built around the minimum geotechnical reporting protocol proposed in this study.

The regional context analysis also clarifies the role of public soil and geoscience data portals. Such sources can support regional enrichment and source characterisation, but they should not be treated as mechanical strength datasets unless raw laboratory UCS or CBR measurements are available.

8. Conclusions

1. The curated dataset is adequate for preliminary decision support but not for universal exact prediction across all stabilised soils.
2. Internal validation substantially overestimated transfer performance compared with grouped-source validation.

3. UCS high-strength classification, CBR pass/fail screening and source-wise ranking are more defensible than exact grouped-source regression for the current data.
4. Synthetic augmentation should remain a training-only sensitivity tool because it produced only modest, task-dependent improvements.
5. Embodied-emissions-aware Pareto screening is a useful practical output because it identifies mixtures that balance performance gain and estimated carbon dioxide equivalent burden.

The contextual-feature comparison confirmed that regional soil-health information produces only modest grouped-source gains. This finding supports a restrained conclusion: physics-constrained and source-aware modelling improves diagnostic reliability, but globally transferable stabilised-soil prediction requires richer specimen-level mineralogical and chemical reporting.

The negative-control comparison shows that regional agricultural descriptors provide marginal mechanical benefit, while the remaining transfer gap confirms that specimen-level mineralogical and chemical descriptors are necessary for a stronger universal model.

CRedit Authorship Contribution Statement

Author contributions should be completed by the authors before submission.

Declaration of Competing Interest

The authors should declare whether they have any competing financial interests or personal relationships that could have influenced the work.

Ethics Statement

This study used previously published literature data and did not involve human participants, animals or field interventions requiring ethical approval.

Data Availability

The reproducibility package includes the curated mechanical dataset, regional contextual table, source inventory, compilation audit, base-versus-context comparison metrics and analysis

scripts. Regional soil-health records are supplied as context only and are not used as additional UCS or CBR measurements.

The cleaned real datasets, source inventory, feature dictionary, analysis scripts, fold assignments, prediction files, augmentation rules and emission-factor table are archived in a private GitHub repository: <https://github.com/bharti19may-ctrl/source-aware-stabilized-soil-ml>. The repository is currently private and can be made available to editors/reviewers on request. Synthetic rows, if released, are stored separately and labelled as training-only auxiliary data.

The curated mechanical dataset, regional contextual table and base-versus-context comparison metrics are archived with the private reproducibility repository. The package includes processed CSV files, source-use decisions and the analysis report.

Appendix A. Additional Sensitivity and Reproducibility Tables

Target	Threshold	Positive class count	Positive class fraction	Purpose
UCS	1200 kPa	880	0.509	High-strength screening sensitivity
UCS	1500 kPa	792	0.458	High-strength screening sensitivity
UCS	1800 kPa	707	0.409	High-strength screening sensitivity
CBR	5%	420	0.866	Bearing-capacity screening sensitivity
CBR	8%	374	0.771	Bearing-capacity screening sensitivity
CBR	10%	319	0.658	Bearing-capacity screening sensitivity

Table A2. Balanced-accuracy drop from internal validation to grouped-source validation.

Task	Internal BA	Grouped-source BA	BA drop	Interpretation
UCS_high_binary	0.934	0.64	0.293	positive value indicates source-transfer loss
UCS_multiclass	0.807	0.342	0.465	positive value indicates source-transfer loss
CBR_pass_8_binary	0.943	0.53	0.413	positive value indicates source-transfer loss
CBR_multiclass	0.744	0.238	0.506	positive value indicates source-transfer loss

Table A1. Threshold sensitivity around the adopted unconfined compressive strength and California bearing ratio screening cut-offs.

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